

DATA SCIENCE 101 — LESSON 03

Welcome to Jupyter

Introduction to Jupyter Notebook: what it is, how to install it with Anaconda, and how to work faster with shortcuts.

Heng Somoradane

What Will We Cover?

1

What is Jupyter Notebook / JupyterLab?

Understand different between **Python** and the **Jupyter Notebook App**.

2

How to use Jupyter

A tour of the **dashboard**, **notebooks**, **cells**, and the difference between **edit** and **command** mode.

3

Shortcuts consumption

Work faster with keyboard shortcuts: **cut/copy/paste cells**, **add/delete cells**, and **markdown** cells.

● **Insight:** Along the way: coding techniques — solving error messages and starting with a clean slate.

WHAT IS JUPYTER

A Server–Client Application

- The **Jupyter Notebook App** (or simply Jupyter) is a **server–client application**.
- You type code on your machine, the **Jupyter server** runs it through **Python**, and the results render right back in your **browser**.



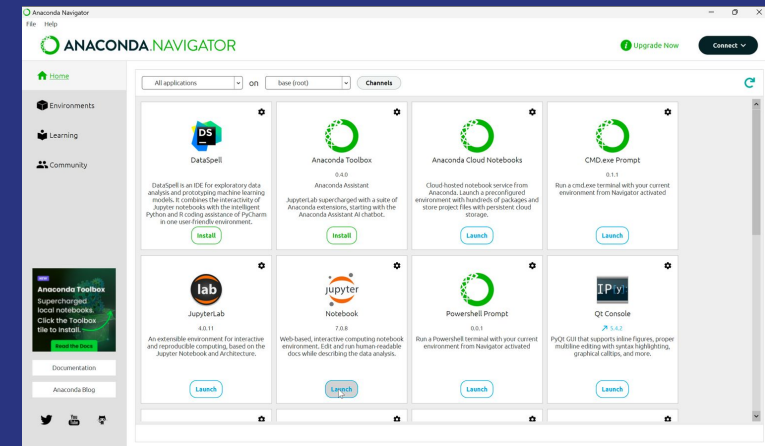
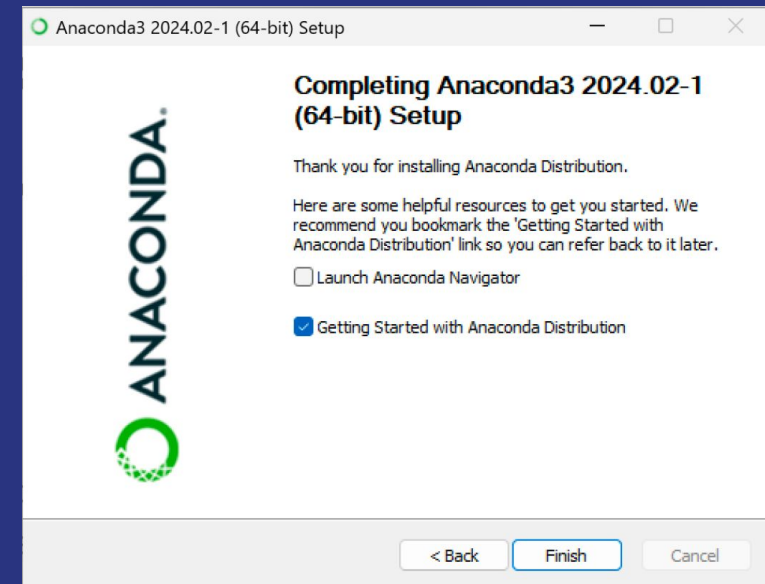
● **Insight:** Notebook files are saved in the IPython Notebook format — *.ipynb for brief.

INSTALLATION — STEP 4 OF 4

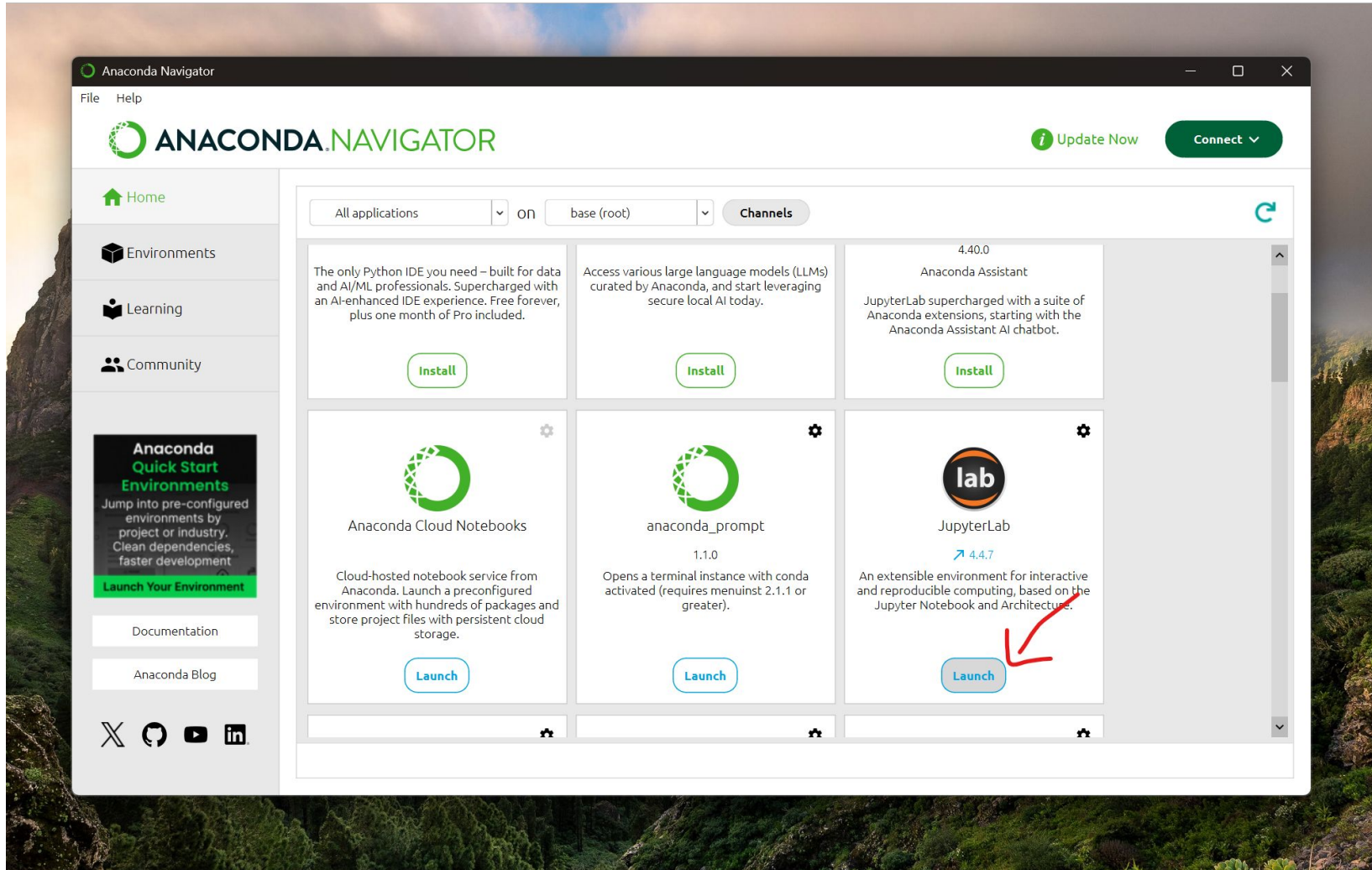
Finish and Launch

Untick the extras.

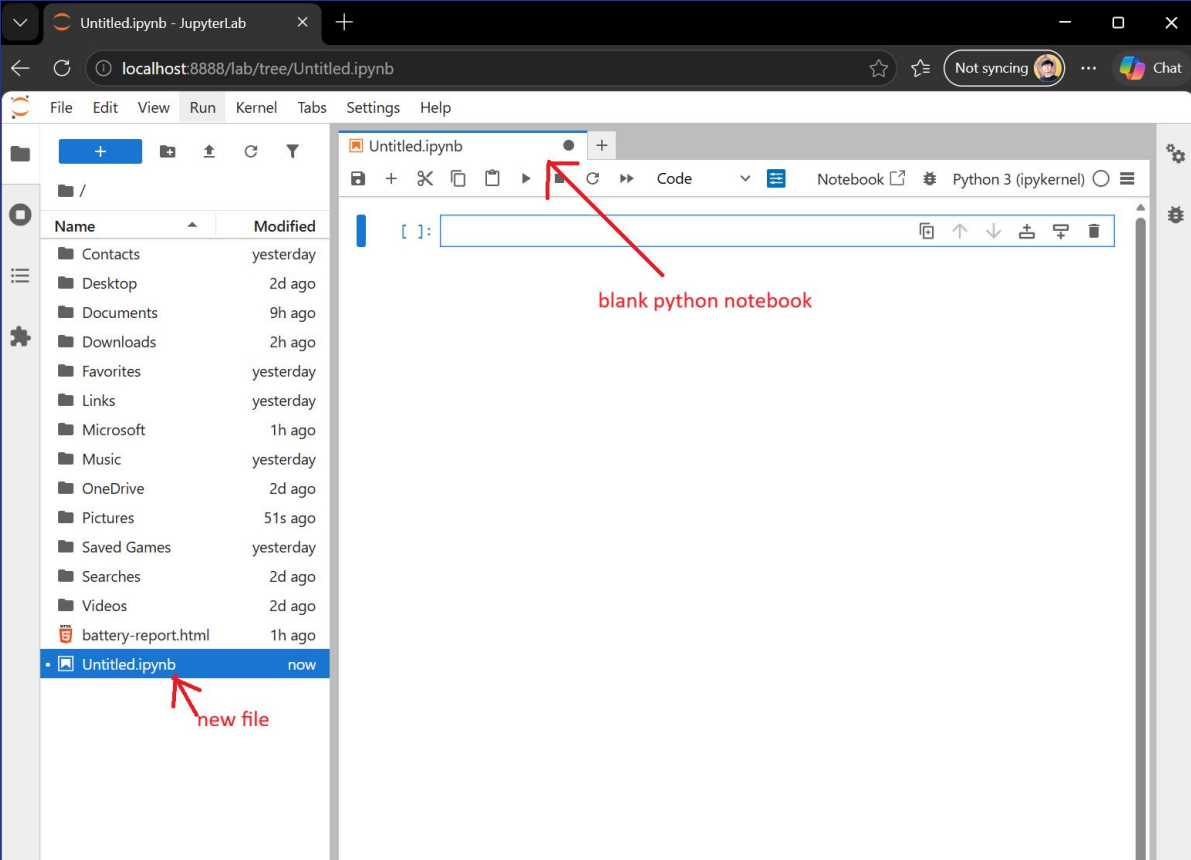
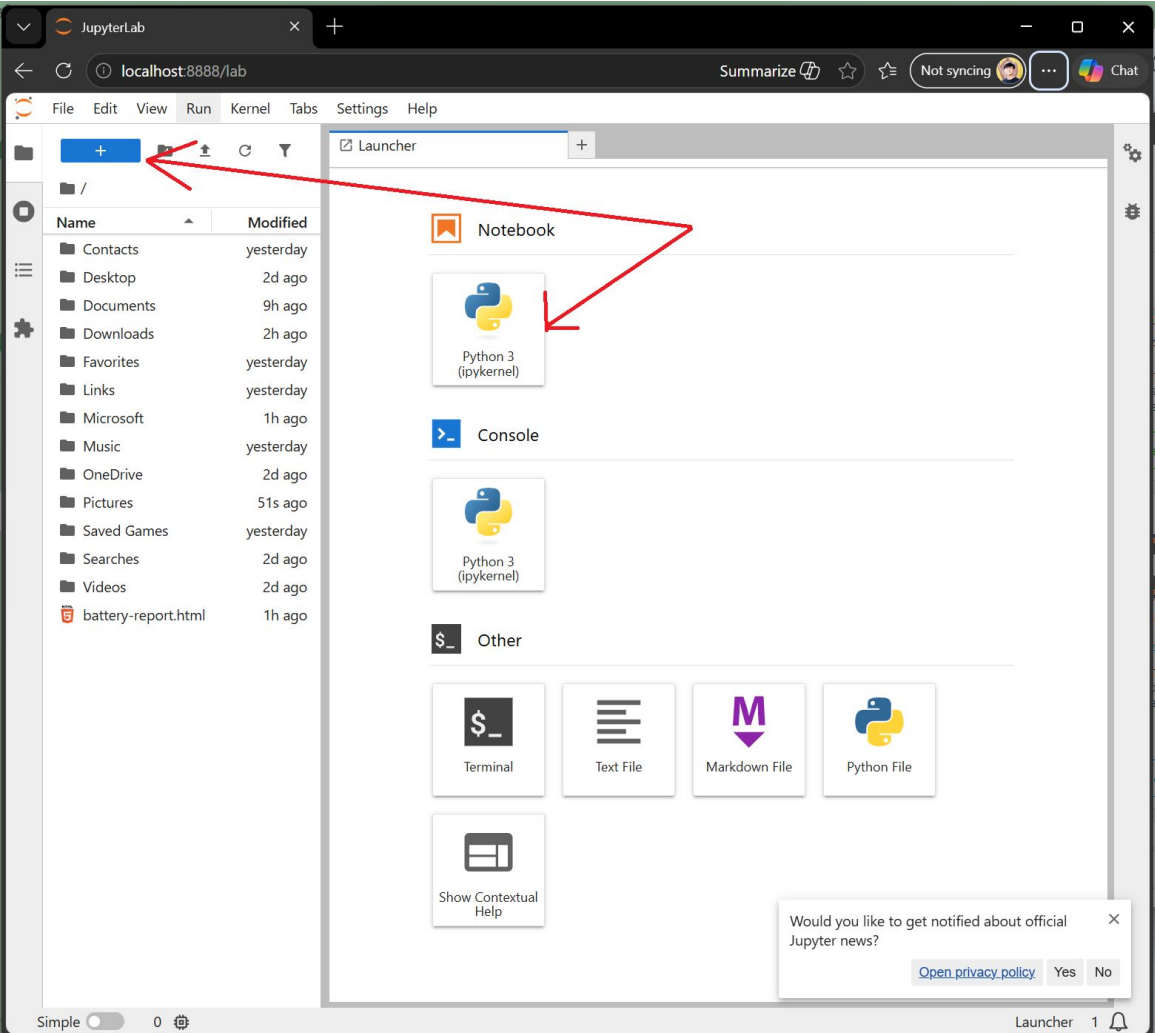
- Click **Next** → **Finish**; you can **untick all selections** on the last screen.
- Find **Anaconda Navigator** in the Start menu and press **Launch** on the **Jupyter Notebook** tile (or open Jupyter Notebook directly).



Open Anaconda



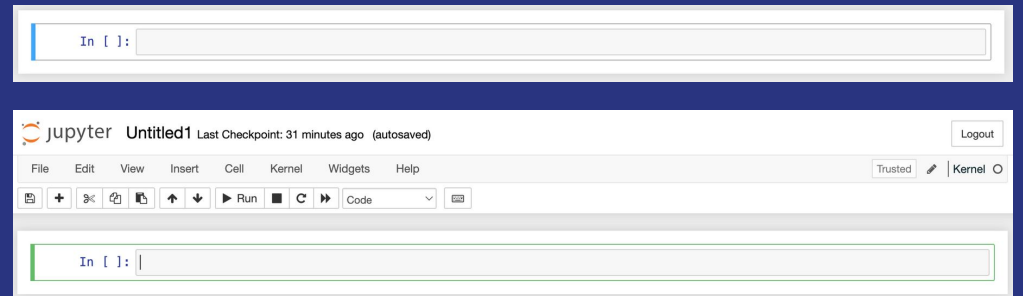
Open and Create Notebook



Cells: Edit vs Command Mode

The gray box is the input field.

- Press **Enter** to enter **edit mode** — a **green border** appears and you can type.
- Press **Escape** to return to **command mode** — the border turns **blue** and keys become shortcuts.



Execute Cells

Ctrl+Enter vs Shift+Enter.

- Type `x = [1, 2, 3, 4]` then `x`, and press **Ctrl + Enter** to **execute in place**.
- Press **Shift + Enter** to execute **and create a new cell** below at the same time.

```
In [1]: x = [1, 2, 3, 4]
        x
Out[1]: [1, 2, 3, 4]
```

```
In [2]: x = [1, 2, 3, 4]
        x
Out[2]: [1, 2, 3, 4]
```

```
In [ ]:
```


Command-Mode Shortcuts

1

Cut, copy, paste cells

X cuts the selected cell, **C** copies it, **V** pastes it below the selected cell.

2

Add cells

A inserts an empty cell **above** the selection; **B** inserts one **below**.

3

Delete cells

Press **D twice** to delete the selected cell — handy when you've made a mistake.

4

Markdown vs code

M turns a cell into **markdown** (documentation, not executed); **Y** turns it back into a **code** cell.

SHORTCUTS IN PRACTICE

Example: Move a Cell with X and V

Select, cut, move, paste.

- Select the **b = a + 2** cell and press **X** to cut it.
- Move up to the target cell and press **V** — the cell is pasted **below** it.
- Only the toolbar **arrow buttons** move cells directly; there's no shortcut for that.



Insight: In command mode, press H to see the full keyboard-shortcuts reference.

```
In [2]: x = [1, 2, 3, 4]
x
Out[2]: [1, 2, 3, 4]

In [ ]:

In [4]: a = 5

In [5]: a
Out[5]: 5

In [6]: b = a + 2
b
Out[6]: 7
```

```
In [2]: x = [1, 2, 3, 4]
x
Out[2]: [1, 2, 3, 4]

In [ ]:

In [4]: a = 5

In [5]: a
Out[5]: 5
```

The Benefits

1

Documentation next to code

When code gets long, **markdown cells** let you leave comments and explain the solution you've created.

2

Run any cell, any time

Select and execute **whichever cell you want** — solve problems in **pieces**.

3

Coding techniques

Read **error messages** (like `NameError`) to fix bugs, and restart the kernel to **start with a clean slate**.

● **Insight:** Practice the shortcuts this week — they quickly become muscle memory.